

INTRODUCTION



This book is about a software development technique called *object-oriented programming (OOP)* and how it can be used with Python. Before OOP, programmers used an approach known as *procedural programming*, also called *structured programming*, which involves building a set of functions (procedures) and passing data around through calls to those functions. The OOP paradigm gives programmers an efficient way to combine code and data into cohesive units that are often highly reusable.

In preparation for writing this book, I extensively researched existing literature and videos, looking specifically at the approaches taken to explain this important and wide-ranging topic. I found that instructors and writers typically start by defining certain key terms: *class*, *instance variable*, *method*, *encapsulation*, *inheritance*, *polymorphism*, and so on.

While these are all important concepts, and I'll cover all of them in depth in this book, I'll begin in a different way: by considering the question, "What problem are we solving?" That is, if OOP is the solution, then what is the problem? To answer this question, I'll start by presenting a few examples of programs built using procedural programming and identifying complications inherent in this style. Then I'll show you how an object-